$$
\begin{aligned}
&C_{u\phi_1}^{(12)}{}^{i1\_i2\_} \to \\
&\frac{1}{16\pi^2}\left(-\frac{1}{4} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{2,2,0}[m_1,\tilde{\mu}] - \frac{1}{12} m_2\tilde{\mu} g_2^4 \left(y_u^{i1i2}+4 y_u^{pi2}\delta_{pi1}\right) \mathrm{LF}_{2,2,0}[m_2,\tilde{\mu}] +\right. \\
&\frac{1}{6} m_2\tilde{\mu} g_2^4 \left(-2 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}\right) \mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}] + \\
&\frac{1}{18}\tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{2,2,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{36}\tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{18}\tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{2,2,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] - \\
&\frac{1}{36}\tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] - \\
&\frac{1}{18}\tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \\
&\frac{1}{18}\tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] + \frac{1}{8}\tilde{\mu}\overline{y_e}^{pr} a_e^{pr} \\
&\quad\left(g_2^2\left(y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \\
&\frac{1}{18}\tilde{\mu}\overline{y_e}^{pr} a_e^{pr} \left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \frac{1}{18}\tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr}\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \\
&\frac{1}{18}\tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \\
&\frac{1}{24}\tilde{\mu}\overline{y_d}^{pr} a_d^{pr} \left(9 g_2^2\left(y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+5 g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \frac{1}{6}\tilde{\mu}\overline{y_d}^{pr} a_d^{pr} \\
&\quad\left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \\
&\frac{1}{36}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \left(9 g_2^2\left(-2 y_u^{i1i2}+y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{3}{4}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{i1i2} a_u^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{72}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \left(9 g_2^2\left(-5 y_u^{i1i2}+y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \frac{3}{4}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{i1i2} a_u^{pr} \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{1}{36}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \\
&\quad\left(9 g_2^2\left(-2 y_u^{i1i2}+y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{36}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \left(9 g_2^2\left(-2 y_u^{i1i2}+y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \frac{1}{12}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \\
&\quad\left(9 g_2^2\left(2 y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{6}\tilde{\mu}\overline{y_u}^{pr} a_u^{pr} \left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{si2}\delta_{si1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{si2}\delta_{si1}-4 y_u^{i1s}\delta_{si2}\right)\right) \\
&\quad\mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \frac{5}{8} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
&\frac{1}{24} m_1\tilde{\mu} g_1^2 \left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{pi2}\delta_{pi1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}-4 y_u^{i1p}\delta_{pi2}\right)\right) \\
&\quad\mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] - \\
&\frac{1}{18} m_1\tilde{\mu} g_1^2 \left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{pi2}\delta_{pi1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}-4 y_u^{i1p}\delta_{pi2}\right)\right) \\
&\quad\mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \frac{1}{3} m_2\tilde{\mu} g_2^4 \left(-2 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}\right) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
&\frac{1}{24} m_2\tilde{\mu} g_2^4 \left(-67 y_u^{i1i2}+8 y_u^{pi2}\delta_{pi1}\right) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\
&\frac{1}{8} m_2\tilde{\mu} g_2^2 \left(7 g_2^2\left(2 y_u^{i1i2}-y_u^{pi2}\delta_{pi1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}-4 y_u^{i1p}\delta_{pi2}\right)\right) \\
&\quad\mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] + 2 m_2\tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] - \\
&\frac{1}{6} m_2\tilde{\mu} g_2^2 \left(3 g_2^2\left(2 y_u^{i1i2}-y_u^{pi2}\delta_{pi1}\right)+g_1^2\left(3 y_u^{i1i2}+y_u^{pi2}\delta_{pi1}-4 y_u^{i1p}\delta_{pi2}\right)\right) \\
&\quad\mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] + \frac{1}{9} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{9} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{1}{12} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \\
&\quad\mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \frac{1}{12} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{36} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{3} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{1}{3} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{4}{9} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] + \\
&\frac{1}{24} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{q}}^{p}\right] - \\
&\frac{11}{72} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}\right] + \frac{11}{24} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}^{r}\right] + \\
&\frac{8}{9} m_1\tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}^{i2}\right] - \frac{3}{4} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{3}{4} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{3}{4} m_2\tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}\left[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{3}{2}\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1+m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
&\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1+m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{1}{4} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \\
&\quad\mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{d}}^{r}\right] + \frac{9}{8} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{8} m_2\tilde{\mu} g_2^4 \left(13 y_u^{i1i2}+2 y_u^{pi2}\delta_{pi1}\right) \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}\right] + \\
&\frac{1}{8} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{u}}^{r}\right] + \\
&\frac{4}{3} m_3\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{4}{3} m_3\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pt} y_u^{i1i2} a_d^{sr} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}\right] - \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{8}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{8}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{sr} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{4}\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1+7 m_2) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
&\frac{3}{2} m_2\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] + \frac{1}{12} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{6} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{3} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{u}}^{r}\right] + \\
&\frac{7}{12} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{u}}^{r}\right] + \frac{1}{2}\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1+3 m_2) \\
&\quad\mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \frac{1}{2} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}\right] - \\
&\frac{3}{4} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}\right] + \frac{3}{2} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \\
&\quad\mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}\right] + \frac{1}{4} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{2}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \frac{1}{2}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \\
&\quad\mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \frac{1}{12} m_1\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{8} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{3}{16} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{4}{9}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] - \\
&\frac{2}{9}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] + \frac{1}{6}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \\
&\quad\mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] - \frac{1}{12}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] + \\
&\frac{1}{36}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \frac{1}{72}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \frac{1}{24}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{1}{48}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{2}{9}\tilde{\mu} g_1^4 a_u^{i1i2} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] - \frac{1}{12} m_2\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{4} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \frac{3}{8} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \frac{1}{2}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{4}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}\right] + \frac{1}{4}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \\
&\quad\mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \frac{1}{8}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{8} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{3}{16} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \frac{1}{4}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{8}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{1}{8}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \\
&\quad\mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \frac{1}{16}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{4}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{1,1,1,1,0}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] - \frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \\
&\quad\mathrm{LF}_{2,1,1,1,-1}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] + \frac{4}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,0}\left[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{8} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{r}, m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{8}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{r}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] + \\
&\frac{1}{2}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st} y_d^{i1t} y_u^{pi2} a_d^{sr} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, \tilde{\mu}\right] + \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st} y_d^{i1t} y_u^{pi2} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, \tilde{\mu}\right] + \\
&\frac{1}{2}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, \tilde{\mu}\right] - \\
&\frac{3}{4}\tilde{\mu}^3 \overline{y_d}^{pr}\overline{y_d}^{st} y_d^{sr} y_u^{i1i2} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st} y_d^{i1t} y_u^{pi2} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, \tilde{\mu}\right] + \\
&\frac{3}{4}\tilde{\mu}^3 \overline{y_d}^{pr}\overline{y_d}^{st} y_d^{sr} y_u^{i1i2} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{4}\tilde{\mu}^3 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{sr} y_u^{i1i2} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] + \\
&\frac{1}{4}\tilde{\mu}^3 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{sr} y_u^{i1i2} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] - \\
&\frac{1}{2}\tilde{\mu}^3 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{sr} y_u^{i1i2} a_e^{pt} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] + \\
&\frac{1}{2}\tilde{\mu}^3 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{sr} y_u^{i1i2} a_e^{pt} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] + \\
&\frac{1}{72}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}\right] + \\
&\frac{5}{48} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{1}{12}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] + \\
&\frac{1}{4}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}\right] - \\
&\frac{1}{8} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_2, m_{\tilde{d}}^{r}, \tilde{\mu}\right] + \\
&\frac{1}{8}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{16} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_2, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, \tilde{\mu}\right] + \\
&\frac{3}{4}\tilde{\mu}^3 \overline{y_d}^{pr} y_d^{sr}\overline{y_u}^{st} y_u^{i1i2} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_d}^{st} y_u^{i1i2} a_d^{pt}\overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{2}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{si2} y_u^{i1r} a_u^{pt} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, \tilde{\mu}\right] + \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{si2} y_u^{i1r} a_u^{pt} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, \tilde{\mu}\right] - \\
&\frac{3}{8}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{8}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{2}\tilde{\mu}^3 \overline{y_d}^{pr}\overline{y_d}^{st} y_d^{sr} y_u^{i1i2} a_d^{pt} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] + \\
&\frac{3}{2}\tilde{\mu}^3 \overline{y_d}^{pr}\overline{y_d}^{st} y_d^{sr} y_u^{i1i2} a_d^{pt} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{si2} y_u^{i1r} a_u^{pt} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, \tilde{\mu}\right] + \\
&\frac{9}{8}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] + \\
&\frac{1}{2}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_u^{sr} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, \tilde{\mu}\right] + \\
&\frac{3}{8}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{72}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{48}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \\
&\frac{1}{4}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{8}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{r}, \tilde{\mu}\right] - \\
&\frac{1}{8}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{16}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \\
&\frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{2}{9}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}\right] - \\
&\frac{5}{48} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{1}{48}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] + \\
&\frac{1}{16} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{16}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}\right] + \\
&\frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}\right] - \\
&\frac{3}{2}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{2}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_u^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, \tilde{\mu}\right] + \\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{i1i2}\overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_u^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \\
&\frac{2}{9}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{i2}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{12}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{i2}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{2}{3}\tilde{\mu} g_3^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{i2}, m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{24}\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (11 m_1-5 m_2) \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}\right] + \\
&\frac{2}{3} m_2\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{u}}^{i2}\right] + \\
&\frac{9}{16} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{4}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}\right] + \\
&\frac{1}{16}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}\right] + \\
&\frac{9}{8} m_2\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{3}{8}\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}\right] + \\
&\frac{9}{16} m_2\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{16}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{r}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st} y_d^{i1t} y_u^{pi2} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_d^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{si2} y_u^{i1r} a_u^{pt} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] - \\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t} a_u^{sr} \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] - \\
&\frac{2}{3}\tilde{\mu} g_1^2 g_2^2 a_u^{i1i2} \mathrm{LF}_{1,1,1,1,1,-1}\left[m_1, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] - \\
&\frac{2}{3}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{i1r} a_u^{pi2} \mathrm{LF}_{1,1,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] + \\
&\frac{1}{6}\tilde{\mu} g_1^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \\
&\frac{2}{9} m_1\tilde{\mu} g_1^2 \overline{y_u}^{pr} a_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] - \\
&\tilde{\mu} g_2^2 \overline{y_d}^{pr} y_u^{pi2} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,-1}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{1}{2}\tilde{\mu} g_2^2 \overline{y_u}^{pr} y_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,-1}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] - \\
&\frac{8}{3} m_3\tilde{\mu} g_3^2 \overline{y_u}^{pr} a_u^{pi2} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}\left[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] - \\
&\left.\tilde{\mu}^3 \overline{y_d}^{pr}\overline{y_d}^{st} y_d^{i1r} y_u^{si2} a_d^{pt} \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, \tilde{\mu}\right]\right)
\end{aligned}
$$